



Problem Statement

Title: Pneumonia Detection from Chest X-ray Images using Convolutional Neural Networks (CNN)

Objective:

Develop an intelligent medical imaging system that uses deep learning (CNN) to **automatically classify chest X-ray images** as either **Pneumonia** or **Normal**. The goal is to support radiologists by offering a fast, accessible, and accurate diagnostic tool.

Motivation:

Pneumonia is a life-threatening illness, especially in children and elderly people. Early detection using automated methods can reduce diagnostic errors, assist in timely treatment, and improve healthcare reach in rural or resource-limited settings.



Dataset Access via KaggleHub

Dataset URL:

<https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>



Directory Structure

pneumonia-cnn-app/

- |— app.py
- |— model_training.ipynb
- |— trained_model.h5
- |— requirements.txt